

2	(a) force per unit positive charge	B1	[1]
(b)	(i) time = $5.9 \times 10^{-2} / 3.7 \times 10^7$ = $1.6 \times 10^{-9} \text{ s}$ ($1.59 \times 10^{-9} \text{ s}$)	A1	[1]
	(ii) $E = V/d$	C1	
	= $2500 / 4.0 \times 10^{-2}$		
	= $6.3 \times 10^4 \text{ NC}^{-1}$ (6.25×10^4 or 62500 NC^{-1})	A1	[2]
	(iii) $a = Eq/m$ or $F = ma$ <u>and</u> $F = Eq$	C1	
	= $(6.3 \times 10^4 \times 1.60 \times 10^{-19}) / 9.11 \times 10^{-31} = 1.1 \times 10^{16} \text{ ms}^{-2}$	A1	[2]
	(iv) $s = ut + \frac{1}{2}at^2$		
	= $\frac{1}{2} \times 1.1 \times 10^{16} \times (1.6 \times 10^{-9})^2$	C1	
	= $1.4 \times 10^{-2} \text{ (m)}$	C1	
	distance from plate = $2.0 - 1.4$ = 0.6 cm (<i>allow 1 or more s.f.</i>)	A1	[3]
	(v) electric force $\gg$ gravitational force (on electron)/weight or acceleration due to electric field $\gg$ acceleration due to gravitational field	B1	[1]
	(vi) v_x-t graph: horizontal line at a non-zero value of v_x	B1	
	v_y-t graph: straight line through the origin with positive gradient	B1	[2]

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